| INCH-POUND |

MIL-A-23987A(YD)
12 January 1989
SUPERSEDING
MIL-A-23987(YD)
06 MAY 1982

MILITARY SPECIFICATION

A-FRAME, PONTOON TUG: PREFABRICATED, KNOCKED DOWN

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and size A-frame used on the bow of pontoon warping tugs for lowering and hoisting anchors.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2020

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SPECIFICATIONS

FEDERAL

FF-T-276 - Thimbles, Rope.
FF-T-791 - Turnbuckle.
RR-W-410 - Wire Rope and Strand.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Nailed and Lock-Corner.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material; Greaseproofed, Waterproofed, Flexible.
MIL-P-24441 - Paint, Epoxy-polyamide, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government drawing. The following other Government drawing form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)

879162 - N.L. Equipment P-Series Warping Tug A-Frame Assembly and Details.

(Copies of drawings can be obtained from CESO (Code 155), CBC, Port Hueneme, CA 93043.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation.

AMERICAN IRON AND STEEL INSTITUTE (AISI)

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Pocketbook of AISI Standard Steels.

(Application for copies should be addressed to the American Iron and Steel Institute, 1000 16th Street NW, Washington, D.C. 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 36 - Structural Steel. (DoD adopted)
ASTM A 53 - Pipe, Steel Black and Hot Dipped Zinc Coated, Welded and Seamless. (DoD adopted)
ASTM A 108 - Steel Bar, Carbon, Cold Finished, Standard Quality.
ASTM D 3951 - Commercial Packaging.
ASTM D 3953 - Strapping, Flat Steel and Seals.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1 - Structural Welding Code - Steel.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, Miami, FL 33135.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 11006, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The A-frame assembly shall be furnished knocked down and shall be made up of prefabricated units consisting of item numbers 1 through 18 as listed in the Bill of Materials shown on drawing 879162.

3.2 Drawings. The drawings forming a part of this specification are engineering design drawings. The contractor is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the items. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.3.1 Structural Steel. Steel plates and bars shall be in accordance with ASTM A 36. Steel pipe shall conform to ASTM A 53.

3.3.2 Sheave Pin. The sheave pin shall be alloy steel equivalent to AISI 4340 or AISI 4140, conforming to the requirements of 3.4.5. Alternately, with the approval of the contracting officer, the contractor may use a suitable case-hardened steel with equivalent tensile properties.

3.4 Construction. The A-frame assembly shall be constructed as shown on drawing 879162, and shall conform to the requirements of this specification.

3.4.1 Leg and crown block units. Legs will be welded to the crown block in field assembly. The proper leg, pad eye, and collar (MK-J) shall be furnished disassembled for welding in the field. The legs shall be of six-inch diameter extra-strong steel pipe of the size and shape shown on the drawing. Plates shall be of structural steel. Load-bearing pins shall be made from steel bar in accordance with ASTM A 108. Cotter pins shall be of corrosion-resistant steel.

3.4.2 Stays. The stays shall be furnished assembled with thimbles and cable clips. Wire rope shall be galvanized and shall conform to RR-W-410, type I, class 2, improved plow steel. Thimbles shall be galvanized conforming to FF-T-276, type III. Cable clips shall be zinc coated and shall have forged saddles. Turnbuckles shall have forged body, jaw and jaw, conforming to FF-T-791, type I, form 1, class 7.

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3.4.3 Pad eye and plate units. Units shall be of structural steel, fabricated and welded, of sizes and shapes shown on the drawing. Holes for pad eyes shall be smooth, round, and of proper diameter.

3.4.4 Fasteners. Bolts, nuts, and washers shall be zinc-coated and of the characteristics, dimensions, and quantities shown on the drawing.

3.4.5 Sheave Assembly. Crown block sheave shall be fabricated in accordance with drawing 879162. Plates and structural shapes shall be of structural steel. Sheave pin shall be fabricated from the material of 3.3 and hardened to Rockwell C 40-42. Case hardened steels shall be hardened to Rockwell C 60-62 for a minimum depth of 1/32 inch.

3.5 Interchangeability. All units of the same classification furnished with similar options shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Cleaning and painting.

3.6.1 Cleaning. All steel surfaces of fabrications to be painted shall be cleaned by blasting. Blast cleaning shall be to a near-white metal blast or cleaner. Near-white metal blast is described as a surface with 95 percent of each square inch free of all visible residue with the remaining 5 percent of each square inch having no more than very light shadows or streaks caused by rust stains or mil scale oxides. Blast cleaning shall utilize either dry sand or crushed grit made of cast iron or steel or synthetic grit. Grit shall be of a maximum particle size no longer than that passing a 16 mesh screen.

3.6.1.1 Prepainted surfaces. Purchased components which are prepainted need not be cleaned by blasting, but shall have accumulated dirt and grease removed. All such surfaces shall be cleaned with water and detergent, rinsed with fresh water and dried, followed by painting as described in 3.6.1.2 through 3.6.1.4.

3.6.1.2 Epoxy polyamide painting. MIL-P-24441 formulations shall be applied only to dry surfaces in an atmosphere having relative humidity not greater than 80 percent, and a temperature of 70 degree Fahrenheit (^{o}F) or higher. When ambient atmosphere does not meet these requirements, all painting shall be accomplished in a controlled humidity environment. There shall be 16 hours of curing between each coat. If more than 7 days elapse between epoxy coats, the surface shall be cleaned with water and detergent, rinsed with fresh water, dried, and then a tack coat (1 to 2 mils wet film thickness) of the last coat applied before application of the next full coat called for. The two components of all the MIL-P-24441 coatings should be mixed in equal volume by first thoroughly stirring each component separately and then stirring them together. After mixing, there should be a waiting period of 2 hours at 50° to 60°F, 1 to 1-1/2 hours at 60° to 70°F, and 1/2 to hour above 70°F, before applying the coating to insure complete curing later. If spray or roller painting is utilized, all corners and welds shall first be thoroughly brush coated for each coat of paint. The paint application shall be uniform in thickness and color, smooth in appearance, free of dry over-spray, pinholes, orange peel, sags, or other film defects. Threaded surfaces, wearing

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surfaces, wires, stays, and pins shall not be painted. All painted areas shall be touched up where paint is removed as a result of handling or assembly at the manufacturer's plant.

3.6.1.3 Primer and intermediate coat. Primer and intermediate coats shall consist of epoxy polyamide paint conforming to MIL-P-24441 and the appropriate detail specification. The primer coat shall be green, formula 150, type 1, conforming to MIL-P-24441/1 and shall have a minimum 2.5 mils dry film thickness. The intermediate coat shall be white, formula 152, type 1, conforming to MIL-P-24441/3 and shall have a minimum 2.5 mils dry film thickness. Priming of cleaning surface shall be completed within a maximum of four hours after blast cleaning. Epoxy polyamide paint shall be applied in accordance with 3.6.2.

3.6.1.4 Finish coat. All exterior surfaces shall be given one finish coat of epoxy polyamide paint, haze grey, formula 151, type 1, conforming to MIL-P-24441/2. The finish coat shall have a minimum 3.0 mils dry film thickness.

3.7 Workmanship.

3.7.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel (see 3.3). Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.7.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers, lockwashers, or cotter pins shall be provided in accordance with drawing 879162, and all bolts, nuts, and screws shall be tight.

3.7.3 Welding. Welding procedures shall be in accordance with AWS D1.1. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the material connected by the welds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality

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program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, and the packaging inspection of 4.4. This inspection shall be performed on each A-frame submitted under this specification.

4.3 Examination. Each A-frame shall be examined for defects listed in table I. Each attribute within each classification of multiple defects shall constitute one defect.

TABLE I. Classification of defects.

Classification	Defects	Requirement paragraph
Major:		
101	Material not as specified and obviously defective or not suited for the purpose.	3.3
102	Construction including welding and fabrication, not as specified.	3.4
103	Leg pad eyes and collars not furnished disassembled, as specified.	3.4.1
	Guard plates not welded to crown block.	
	Legs not fabricated of proper size and type of steel pipe.	
Major:		
104	Stays not furnished assembled with thimbles and cable clips as specified.	3.4.2
	Wire rope, cable clips, and turnbuckles not type specified.	
105	Pad eye and plate units not fabricated and welded as specified.	3.4.3
106	Fasteners not zinc coated or of the characteristics, dimensions, and quantities	3.4.4

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specified.

TABLE I. Classification of defects (continued).

Classification	Defects	Requirement paragraph
107	Component parts not interchangeable, and critical dimensions and tolerances not as specified.	3.5
108	Cleaning, pretreatment, prime coating, intermediate coating, finish coating, film thickness, and general paint application not as specified.	3.6
109	Workmanship inferior and not as specified.	3.7

4.4 Packaging inspection. The preservation, packing, and marking of the A-frames and components shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. The preservation shall be level A, or commercial, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Disassembly. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss and to accomplish reduction in cube. Bolts, nuts, pins, and washers removed shall be reinstalled in one of the mating parts.

5.1.1.2 Preservatives. Preservatives shall conform to the applicable specifications listed in, and shall be applied in accordance with MIL-P-116.

5.1.1.3 Unpainted surfaces. Unpainted, uncoated, exposed ferrous metal surfaces of bolts, nuts, including threaded surfaces and pulley grooves shall be coated with type P-1 preservative.

5.1.1.4 Wire rope. Wire rope shall be preserved method I by coating with type P-3 preservative. The preservative shall be applied to all surfaces of the wire rope. Excess preservative shall be allowed to dry (at least 4 hours). The wire rope shall be bundled or coiled and wrapped with a barrier material conforming to MIL-B-121, type I, grade A, class 2 or grade C and secured.

5.1.2 Commercial. The complete A-frame assembly shall be preserved to afford protection against deterioration and damage as required by

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ASTM D 3951. This shall include, but not be limited to, the disassembly requirements in 5.1.1.1 and the preservation, coiling, and securing of wire rope, as in 5.1.1.4

5.2 Packing. Packing shall be level A, B, or commercial, as specified (see 6.1).

5.2.1 Level A. The leg and crown block units for each A-frame assembly shall be nested, arranged, and secured with steel strapping in combination with suitable wood blocking and battens as required to form compact non-shifting bundles. Strapping shall be 0.035 by 1-1/4-inch flat steel conforming to ASTM D 3953, class 1, type I, finish B. Strapping shall be stapled to any wood blocking and battens provided. The stays, pad eye, plate units, fasteners, and any other unassembled components shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. The contents shall be blocked and braced to prevent movement within the boxes.

5.2.2 Level B. The A-frame assemblies shall be packed as specified for level A, except that the boxes shall be domestic type or class, as applicable.

5.2.3 Commercial. The A-frame assemblies shall be packed in bundles and containers in a manner to insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules and shall be in accordance with ASTM D 3591.

5.3 Marking.

5.3.1 Military agencies. Shipments to military agencies shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. A-frames covered by this specification are intended for use on the bow of pontoon warping tugs for lowering and hoisting anchors.

6.2 Acquisition requirements. Acquisition documents should specify the following.

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Level of preservation, and level of packing required (see 5.1 and 5.2).
d. When first article is required for inspection and approval (see 3.1).

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6.3 Subject term (key word) listing.

Anchors
Crown block
Hoist
Sheave
Stay

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy - YD

Preparing activity
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-23987A(YD)	MILITARY SPECIFICATION - A-FRAME, PONTOON TUG: PREFABRICATED, KNOCKED DOWN

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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